

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 12: Friday 18 September 2026

(Week 6)

Remark. Q11.2 to Q11.4 are leftover exercises from the Tuesday class.

Remark. Q012.1 to Q12.2 are also on Section 13.8.

The method of Lagrange multipliers

Goal: To maximize/minimize the function $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$ subject to the constraint

$$\underline{g(\mathbf{x}, \mathbf{y}, \mathbf{z}) = c.} \quad (1)$$

(Here c is a given constant.)

If $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$ attains max or min at a point $P(\mathbf{x}, \mathbf{y}, \mathbf{z})$ subject to the constraint in (1), then there exists some real number λ such that

$$\nabla f(P) = \lambda \nabla g(P).$$

constrained
critical
points

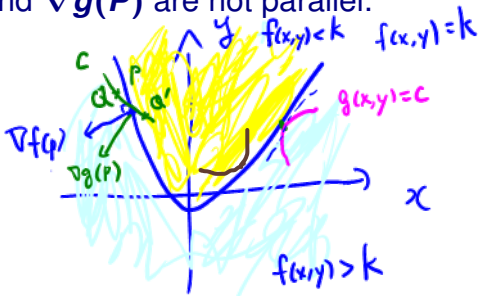
(In other words, $\nabla f(P)$ and $\nabla g(P)$ are parallel.)

Remark. Similar statements hold for functions of two variables x, y .

Graphical illustration: Consider the function in Q06.3.5:

$$\underline{f(x, y) = x^2 - y.}$$

Let $\mathbf{C}$ be the curve defined by the constraint $g(x, y) = c$. Suppose $\mathbf{P}$ is a point on $\mathbf{C}$ such that $\mathbf{f}(\mathbf{P}) = k$, but $\nabla \mathbf{f}(\mathbf{P})$ and $\nabla \mathbf{g}(\mathbf{P})$ are not parallel.



Then one sees that there are points $\mathbf{Q}, \mathbf{Q}'$ on $\mathbf{C}$ near $\mathbf{P}$ where $\mathbf{f}(\mathbf{Q}) > k$ and $\mathbf{f}(\mathbf{Q}') < k$.

Thus $f(\mathbf{x}, \mathbf{y})$ cannot attain max or min at $\mathbf{P}$ subject to the constraint $g(\mathbf{x}, \mathbf{y}) = c$.

In other words, if $f(\mathbf{x}, \mathbf{y})$ attains max or min at $\mathbf{P}$ subject to the constraint $g(\mathbf{x}, \mathbf{y}) = c$, then $\nabla f(\mathbf{P})$ and $\nabla g(\mathbf{P})$ must be parallel. So there exists some number λ such that

$$\nabla f(\mathbf{P}) = \lambda \nabla g(\mathbf{P}).$$

This leads to the **method of Lagrange multipliers** to solve the following optimization problem:

Problem: To maximize/minimize the function $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$ subject to the constraint

$$g(\mathbf{x}, \mathbf{y}, \mathbf{z}) = c. \quad (1)$$

(Here c is a given constant.)

Solution/Method: Step 1: Find 'critical points' of f subject to the constraint in (1) by solving the following system of equations:

$$(*) \quad \begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z), \\ g(x, y, z) = c. \end{cases}$$

Note that the unknowns in (*) are x, y, z, λ .

Step 2: If there are additional 'boundary conditions', then

- (a) find candidate max/min points of f along the boundary.
- (b) discard those solutions in Step 1 that do not satisfy the 'boundary conditions'.

Step 3: Compute and compare the values of f at the candidate max/min points arising from Step 1 and Step 2.

Q11.1. Use the method of Lagrange multipliers to find the maximum value of the function $f(\mathbf{x}, \mathbf{y}) = \mathbf{xy}$ subject to the constraint

$$(\mathbf{x} - 2)^2 + \mathbf{y}^2 = 4.$$

Solution: To maximize $f(x,y) = xy$
Subject to constraint $(x-2)^2 + y^2 = 4$

Let $g(x,y) = (x-2)^2 + y^2$ $g(x,y) = 4$.

Step: $\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$, $\nabla g = \begin{pmatrix} g_x \\ g_y \end{pmatrix} = \begin{pmatrix} 2(x-2) \\ 2y \end{pmatrix}$

We consider the system of equations:

Unknowns: x, y, λ $\begin{cases} \nabla f(x,y) = \lambda \nabla g(x,y) & \Leftrightarrow \begin{cases} y = \lambda \cdot 2(x-2) & \text{--- (1)} \\ x = \lambda \cdot 2y & \text{--- (2)} \\ g(x,y) = 4 & \text{--- (3)} \end{cases} \end{cases}$

(1) $\div$ (2) gives $\frac{y}{x} = \frac{\cancel{2}(x-2)x}{\cancel{2}yx} \Rightarrow y^2 = x^2 - 2x$ --- (4)

Subs (4) into (3), $(x-2)^2 + x^2 - 2x = 4$
 $2x^2 - 6x = 0$

$$2x^2 - 6x = 0$$

$$2x(x-3) = 0$$

$$\Rightarrow x = 0 \text{ or } x = 3$$

by (1)

Case 1: By (4), When $x = 0$, $y^2 = 0^2 - 2 \cdot 0 = 0 \Rightarrow y = 0 \Rightarrow \lambda = 0$
 $(x, y, \lambda) = (0, 0, 0)$.

Case 2: $x = 3$ by (4), $y^2 = 3^2 - 6 = 3 \Rightarrow y = \sqrt{3}, -\sqrt{3}$
 $y = \sqrt{3} \Rightarrow \lambda = \frac{\sqrt{3}}{2}$.

Case 2(a) $(x, y, \lambda) = (3, \sqrt{3}, \frac{\sqrt{3}}{2})$

Case 2(b), $y = -\sqrt{3}$, $\lambda = -\frac{\sqrt{3}}{2}$, $(x, y, \lambda) = (3, -\sqrt{3}, -\frac{\sqrt{3}}{2})$.

Check:- The 3 solutions satisfy (1), (2), (3).

Step 3: $f(0,0) = 0 \cdot 0 = 0$

$$f(3, \sqrt{3}) = 3\sqrt{3}$$

$$f(3, -\sqrt{3}) = -3\sqrt{3}$$

f attains constrained maximum value $3\sqrt{3}$,
at the point $(3, \sqrt{3})$.

Redo the steps:

$$\text{Step 1: } \begin{cases} y = \lambda^2(x-2) & \text{--- (1)} \\ x = \lambda^2 y & \text{--- (2)} \\ (x-2)^2 + y^2 = 4 & \text{--- (3)} \end{cases} \quad \text{unknowns: } x, y, \lambda$$

Case 1: $x=0$. Then by (3), $(0-2)^2 + y^2 = 4$

$$\Rightarrow y^2 = 0 \Rightarrow y = 0$$

Then by (1), $0 = \lambda \cdot 2 \cdot (0-2) \Rightarrow \lambda = 0$

$$(x, y, \lambda) = (0, 0, 0). \quad (4)$$

Then one checks that (4) satisfies (1), (2), (3).

Case (2): $x \neq 0$. Then by (2), $0 \neq x = \lambda \cdot 2y \Rightarrow y \neq 0, \lambda \neq 0$.

Use Tuesday's calculation to get

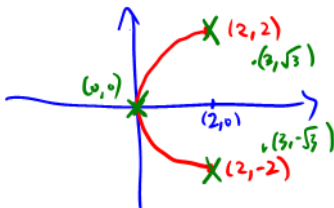
$$(x, y, z) = (3, \sqrt{3}, \frac{\sqrt{3}}{2}), (3, -\sqrt{3}, -\frac{\sqrt{3}}{2}).$$

Step 3: same as before.

Q11.2. Use the method of Lagrange multipliers to find the maximum value of the function $f(\mathbf{x}, \mathbf{y}) = \mathbf{xy}$ subject to the constraints

$$(\mathbf{x} - 2)^2 + \mathbf{y}^2 = 4 \quad \text{and} \quad \underline{\mathbf{x} \leq 2}. \quad (*)$$

[Remark. The above question is modified from Q11.1 by adding an additional constraint $\mathbf{x} \leq 2$. So you may find your calculations in Q11.1 useful.]



$$x \leq 2$$

Solution: The constraint set consists of the 'interior points' given by

$$(x - 2)^2 + y^2 = 4 \quad \text{and} \quad x < 2, \quad (1)$$

as well as the 'boundary points' given by

Step 2:

$$(x - 2)^2 + y^2 = 4 \quad \text{and} \quad x = 2$$
$$\iff (x, y) = (\underline{2}, \underline{2}), (\underline{2}, \underline{-2}). \quad (2)$$

We first find the 'critical points' of $f(x, y)$ subject to the constraints in (1). In turn, we first find the 'critical points' of $f(x, y)$ subject to the constraint $(x - 2)^2 + y^2 = 4$.

This was done in Q11.1 by using the method of Lagrange multipliers, and such points are

Step 1: $(\underline{0}, 0), (3, \underline{\sqrt{3}}), (3, -\underline{\sqrt{3}}).$

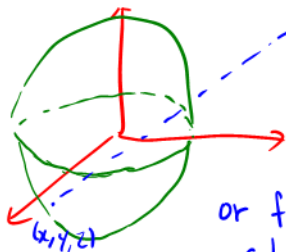
Only $(0, 0)$ satisfies the inequality $x < 2$ in (1). Thus the candidate maximum points of f satisfying (1) is $(0, 0)$.

The candidate maximum points of f satisfying (2) are $(2, 2), (2, -2).$

Step 3: Now we have $f(0, 0) = 0 \cdot 0 = 0,$
 $f(2, 2) = \underline{2 \cdot 2 = 4},$
 $f(-2, 2) = -2 \cdot 2 = -4.$

Hence the maximum value of $f(x, y)$ subject to the constraints in (*) is $\underline{4}$, which is attained at the point $(2, 2).$

Q11.3. Find the point(s) on the sphere $x^2 + y^2 + z^2 = 1$ furthest from the point $(2, 1, 2)$.



$(2, 1, 2)$

Objective : To maximize

$$\sqrt{(x-2)^2 + (y-1)^2 + (z-2)^2}$$

or $f(x, y, z) = (x-2)^2 + (y-1)^2 + (z-2)^2$
subject to the

$$\text{constraint : } x^2 + y^2 + z^2 = 1$$

Solution: The distance of a point (x, y, z) from $(2, 1, 2)$ is

$$\sqrt{(x - 2)^2 + (y - 1)^2 + (z - 2)^2}.$$

Consider the functions

$$f(x, y, z) = (x - 2)^2 + (y - 1)^2 + (z - 2)^2,$$

$$g(x, y, z) = x^2 + y^2 + z^2.$$

Since $\sqrt{f(x, y, z)}$ increases with $f(x, y, z)$, our goal is:

Maximize $f(x, y, z)$ subject to the constraint

$$g(x, y, z) = 1.$$

We use the method of Lagrange multipliers. We have

$$\nabla f(x, y, z) = \begin{pmatrix} 2(x - 2) \\ 2(y - 1) \\ 2(z - 2) \end{pmatrix}, \quad \nabla g(x, y, z) = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix}.$$

We consider the system of equations:

$$\begin{aligned} & \begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z), \\ g(x, y, z) = 1 \end{cases} \quad \text{unknowns: } x, y, z, \lambda \\ \Leftrightarrow & \begin{cases} \cancel{2}(x - 2) = \lambda \cdot \cancel{2}x, \\ \cancel{2}(y - 1) = \lambda \cdot \cancel{2}y, \\ \cancel{2}(z - 2) = \lambda \cdot \cancel{2}z, \\ x^2 + y^2 + z^2 = 1 \end{cases} \quad (*) \\ \Rightarrow & \begin{cases} x(1 - \lambda) = 2, \\ y(1 - \lambda) = 1, \\ z(1 - \lambda) = 2, \\ x^2 + y^2 + z^2 = 1 \end{cases} \quad \Rightarrow 1 - \lambda \neq 0 \end{aligned}$$

$$\Rightarrow \begin{cases} x = \frac{2}{1-\lambda}, & (1) \\ y = \frac{1}{1-\lambda}, & (2) \\ z = \frac{2}{1-\lambda}, & (3) \\ x^2 + y^2 + z^2 = 1. & (4) \end{cases} \quad \lambda = \frac{2}{1-(-2)} = \frac{2}{3}$$

Substituting (1), (2), (3) into (4), we get

$$\begin{aligned} & \frac{2^2}{(1-\lambda)^2} + \frac{1^2}{(1-\lambda)^2} + \frac{2^2}{(1-\lambda)^2} = 1 \\ \Rightarrow & 9 = (1-\lambda)^2 \\ \Rightarrow & 1-\lambda = 3, -3 \\ \Rightarrow & \lambda = -2, 4. \end{aligned} \quad \begin{aligned} & \lambda^2 - 2\lambda - 8 = 0 \\ & (\lambda-4)(\lambda+2) = 0 \end{aligned}$$

When $\lambda = -2$, it follows from (1), (2), (3) that

$$(x, y, z) = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right), \quad \underline{\lambda = -2}. \quad (5)$$

When $\lambda = 4$, it follows from (1), (2), (3) that

$$(x, y, z) = \left(-\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right), \quad \underline{\lambda = 4}. \quad (6)$$

Now one checks that (5) and (6) indeed satisfy (*).

Thus the solution of (*) are given by (5) and (6).

Note that

$$\begin{aligned} f\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right) &= \left(\frac{2}{3} - 2\right)^2 + \left(\frac{1}{3} - 1\right)^2 + \left(\frac{2}{3} - 2\right)^2 \\ &= \frac{16}{9} + \frac{4}{9} + \frac{16}{9} = \underline{4.} \end{aligned}$$

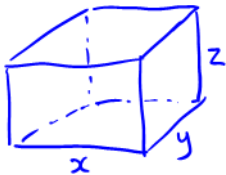
$$\begin{aligned} f\left(\underline{-\frac{2}{3}}, \underline{-\frac{1}{3}}, \underline{-\frac{2}{3}}\right) &= \left(-\frac{2}{3} - 2\right)^2 + \left(-\frac{1}{3} - 1\right)^2 + \left(-\frac{2}{3} - 2\right)^2 \\ &= \frac{64}{9} + \frac{16}{9} + \frac{64}{9} = \underline{16.} \end{aligned}$$

Thus the point on $x^2 + y^2 + z^2 = 1$ furthest from $(2, 1, 2)$ is $\left(-\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right)$ with the distance equal to

$$\sqrt{f\left(-\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right)} = \sqrt{16} = \underline{\underline{4.}}$$

Q11.4. (Q08.5. revisited) A rectangular box with its top left open has its five faces made from different materials. The front face and the back face are made from materials costing 10 dollars per square meter. The two lateral faces are made from materials costing 5 dollars per square meter. The bottom face is made from materials costing 30 dollars per square meter. Using the method of Lagrange multipliers, find the minimum material cost of the rectangular box if the volume of the box is 6 cubic meters. Give also the dimensions (i.e. length, width, and height) of such box.

Solution: Let x , y , z be the length, width and height (in meters) of the rectangular box, and let f be the corresponding material cost (in dollars) of the box. Then



$$\begin{aligned} f(x, y, z) &= 2 \cdot 10 \cdot xz + 2 \cdot 5 \cdot yz + 30 \cdot xy \\ &= 20xz + 10yz + 30xy. \end{aligned}$$

Since volume of the box is **6**, we have the constraints

$$g(x, y, z) = xyz = 6, \quad x > 0, \quad y > 0, \quad z > 0. \quad (*)$$

Our goal is to minimize f subject to the constraints in (*).

We use the method of Lagrange multipliers. We have

$$f(x, y, z) = 20xz + 10yz + 30xy \implies \nabla f = \begin{pmatrix} 20z + 30y \\ 10z + 30x \\ 20x + 10y \end{pmatrix},$$

$$g(x, y, z) = xyz \implies \nabla g = \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}.$$

To minimize f subject to the constraint $g(x, y, z) = 6$, we consider the system of equations

unknowns: x, y, z, λ

$$\begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) \\ g(x, y, z) = 6 \end{cases} \iff \begin{cases} 20z + 30y = \lambda yz \\ 10z + 30x = \lambda xz \\ 20x + 10y = \lambda xy \\ xyz = 6 \end{cases}$$

3 linear equations
in $\frac{1}{y}, \frac{1}{z}, \frac{1}{x}$

$$\Leftrightarrow \begin{cases} \frac{20}{y} + \frac{30}{z} = \lambda & (1) \\ \frac{10}{y} + \frac{30}{z} = \lambda & (2) \\ \frac{20}{y} + \frac{10}{x} = \lambda & (3) \\ xyz = 6. & (4) \end{cases}$$

Here we note that (4) implies that $x, y, z \neq 0$.

Now (1)-(2) gives $\Rightarrow 20x - 10y = 0 \Rightarrow y = 2x$

$$\frac{20}{y} - \frac{10}{x} = 0 \Rightarrow \underline{y = 2x}. \quad (5)$$

(1)-(3) gives

$$\frac{30}{z} - \frac{10}{x} = 0 \Rightarrow \underline{z = 3x}. \quad (6)$$

Substituting (5) and (6) into (4), we get

$$x(2x)(3x) = 6 \Rightarrow x^3 = 1 \Rightarrow x = 1.$$

Together with (5), (6) and (1), we get

$$\mathbf{x} = \mathbf{1}, \mathbf{y} = \mathbf{2}, \mathbf{z} = \mathbf{3}, \lambda = \frac{\mathbf{20}}{\mathbf{2}} + \frac{\mathbf{30}}{\mathbf{3}} = \mathbf{20}. \quad (7)$$

A direct check shows that (7) satisfies (1), (2), (3), (4) and well as the inequalities in (*).

Thus the minimum material cost is attained when $\mathbf{x} = \mathbf{1}$, $\mathbf{y} = \mathbf{2}$, $\mathbf{z} = \mathbf{3}$, and the minimum material cost is

$$f(\mathbf{1}, \mathbf{2}, \mathbf{3}) = \mathbf{20} \cdot \mathbf{1} \cdot \mathbf{3} + \mathbf{10} \cdot \mathbf{2} \cdot \mathbf{3} + \mathbf{30} \cdot \mathbf{1} \cdot \mathbf{2} = \mathbf{180}.$$

$$x, y, z \geq 0$$

Q12.1. A rectangular box is placed in the first octant so that one corner **Q** is at the origin and the three sides adjacent to **Q** lie in the coordinate planes. The corner **P** diagonally opposite to **Q** lies on the ellipsoid

$$\frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4} = 1.$$

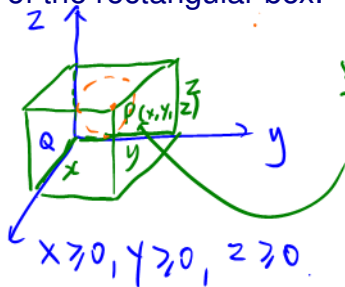
Use the method of Lagrange multipliers to determine the largest possible volume of the rectangular box.

Objective: To maximize

$$V(x, y, z) = xyz$$

subject to the constraint

$$\frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4} = 1,$$



$$\frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4} = 1$$

Solution: Let the point $\mathbf{P}$ be (x, y, z) . Then the volume of the rectangular box is

$$V(x, y, z) = xyz.$$

Note that $\mathbf{P}$ lies on the first octant, so that $x, y, z \geq 0$.

Furthermore, if any of x, y, z is 0 , then $V = 0$. Since our objective is to maximize V , we must assume that

$x, y, z > 0$. Note that $\mathbf{P}$ lies on the given ellipsoid. Let

$$g(x, y, z) = \frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4}.$$

Thus our objective is to:

Maximize $V(x, y, z) = xyz$ subject to the constraints

$$g(x, y, z) = 1, \quad x > 0, y > 0, z > 0.$$

We use the method of Lagrange multipliers. We have

$$V(x, y, z) = xyz, \quad g(x, y, z) = \frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4}. \text{ So}$$

$$\nabla V(x, y, z) = \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}, \quad \nabla g(x, y, z) = \begin{pmatrix} \frac{2x}{3} \\ \frac{2y}{9} \\ \frac{2z}{4} \end{pmatrix}.$$

We consider the system of equations:

$$\Rightarrow \begin{cases} \nabla V(x, y, z) = \lambda \nabla g(x, y, z), \\ g(x, y, z) = 1 \end{cases}$$

$$\Rightarrow \begin{cases} yz = \lambda \cdot \frac{2x}{3}, & (1) \\ xz = \lambda \cdot \frac{2y}{9}, & (2) \\ xy = \lambda \cdot \frac{2z}{2}, & (3) \\ \frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4} = 1. & (4) \end{cases}$$

$$\frac{y}{x} = \frac{3x}{y}$$

unknowns:
 x, y, z, λ .

Since $x, y, z > 0$, it follows from (1) that $\lambda > 0$.

Now (1) $\div$ (2) gives

$$\frac{y}{x} = \frac{3x}{y} \implies y^2 = 3x^2 \implies y = \sqrt{3}x. \quad (5)$$

Also (1) $\div$ (3) gives

$$\frac{z}{x} = \frac{4x}{3z} \implies z^2 = \frac{4}{3}x^2. \implies z = \frac{2}{\sqrt{3}}x. \quad (6)$$

Substituting (5) and (6) into (4), we get

$$\frac{x^2}{3} + \frac{3x^2}{9} + \frac{4}{3} \cdot \frac{x^2}{4} = 1 \implies x^2 = 1 \implies x = 1.$$

Together with (5) and (6), we get $y = \sqrt{3}$, $z = \frac{2}{\sqrt{3}}$.

Together with (1), we get

$$\lambda = \frac{3yz}{2x} = 3 \cdot \sqrt{3} \cdot \frac{2}{\sqrt{3}} \cdot \frac{1}{2} = \underline{3}.$$

Then one checks that $(x, y, z, \lambda) = (1, \sqrt{3}, \frac{2}{\sqrt{3}}, 3)$
satisfies (1), (2), (3), (4).

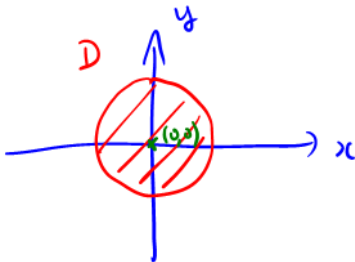
Hence the maximum volume is:

$$V(1, \sqrt{3}, \frac{2}{\sqrt{3}}) = 1 \cdot \sqrt{3} \cdot \frac{2}{\sqrt{3}} = \underline{\underline{2}}.$$

Q12.2. Let $f(x, y) = x^2 - y^2$ be defined on the $D := \{(x, y) : x^2 + y^2 \leq 1\}$.

(a) Find the critical points of f .

(b) What are the maxima and minima of f on D ?



Boundary of D :

$$x^2 + y^2 = 1$$

Solution: (a) Given that $f(x, y) = x^2 - y^2$. Thus

$$f_x(x, y) = 2x, \quad f_y(x, y) = -2y.$$

To find the critical points of f , we solve

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \implies \begin{cases} 2x = 0 \\ -2y = 0 \end{cases} \implies x = 0, y = 0.$$

Note that $(0, 0) \in D$, since $0^2 + 0^2 = 0 \leq 1$.

Hence f has a critical point $(0, 0) \in D$.

(b) From part (a), f has a critical point in D at $(0, 0)$.

We also need to find the extrema of f along the boundary ∂D .

We use the method of the Lagrange multipliers.

Consider the function $g(x, y) = x^2 + y^2$. Then ∂D is the 1-level curve of g . We have

$f(x, y) = x^2 - y^2$
 $g(x, y) = x^2 + y^2$

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ -2y \end{pmatrix}, \quad \nabla g(x, y) = \begin{pmatrix} 2x \\ 2y \end{pmatrix}.$$

To find the critical points of f along ∂D , we consider

unknowns: x, y, λ

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = 1 \end{cases} \implies \begin{cases} 2x = 2\lambda x & (1) \\ -2y = 2\lambda y & (2) \\ x^2 + y^2 = 1 & (3) \end{cases}$$

From (1), we have $2x = 2\lambda x \Leftrightarrow 2x(1-\lambda) = 0$

$$2x(1 - \lambda) = 0 \implies x = 0 \text{ or } \lambda = 1.$$

Case (i): $x = 0$. Substituting this into (3), we get

$$0^2 + y^2 = 1 \implies y = \pm 1.$$

Then from (2), we have

$$y \neq 0 \implies \lambda = -1.$$

So we get two solutions:

$$(x, y, \lambda) = (0, 1, -1), (0, -1, -1).$$

One checks that both solutions satisfy (1), (2), (3).

Cae (ii): $\lambda = 1$. Substituting this into (2), we get

$$-2y = 2 \cdot 1 \cdot y \implies y = 0.$$

Then by (3), we have

$$x^2 + 0^2 = 1 \implies x = \pm 1.$$

Hence we get two solutions:

$$(x, y, \lambda) = (1, 0, 1), (-1, 0, 1).$$

One checks that both solutions satisfy (1), (2), (3).

Thus there are 4 constrained critical points of f :

$$(0, 1), (0, -1), (1, 0), (-1, 0).$$

Recall from part (a) that f has a critical point on D given by $(0, 0)$.

By a direct calculation, we have

$$\begin{aligned} f(0, 0) &= 0, & f(0, 1) &= -1, & f(0, -1) &= -1, \\ f(1, 0) &= 1, & f(-1, 0) &= 1. \end{aligned}$$

Thus the maxima of f on D are $(1, 0)$ and $(-1, 0)$, and corresponding maximum value of f is 1..

The minima of f on D are $(0, 1)$ and $(0, -1)$, and corresponding minimum value of f is -1 .